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WARNING

A BADLY MAINTAINED CALORIMETER IS A MAJOR HAZARD

USE ONLY SILICONE GREASE ON SEALS.

NEVER IMMERSE THE BOMBS IN WATER

OVERVIEW

BOMB

The DRYCAL Calorimeter System utilises a dry bomb, comprising a stainless steel combustion vessel, onto which a tightly fitting thermally conductive jacket is shrunk. The temperature sensors that measure Bomb temperature are cast into this jacket.

Oxygen charging and discharging is controlled by a Valvepin that serves to prevent accidental opening of the bomb while it is pressurised. A lever on the Valvepin moves through a gated Locking Block to occupy the different positions required for charging and discharging the Bomb with Oxygen

Each Bomb is supplied with it's own matching Bombcap, both bear the same Serial Number. They must be used in conjunction with each other, Bombcaps must not be interchanged with different Bombs

The bomb ID for each Bomb/Bombcap assembly is coded into the plastic handle, the number being stamped onto the handle. The handle can thus be moved from one Bombcap to another if it is required to alter the Bomb's ID. Bomb IDs from 1 to 8 are catered for.

CALORIMETER and CONSTANT TEMPERATURE TANK

Water from the thermostatically controlled Constant Temperature Tank is pumped through passages on the walls of the Calorimeter Outer Jacket to provide a stabilised temperature environment around the Bomb during the determination.

OXYGEN FILLING STATION

Oxygen is filtered and regulated to 2500 KPA. A tap controls flow to the Filler Coupling and thus into the Bomb.

BOMB COOLING AND WASHING STATION

The Bomb is placed inverted on this unit to accomplish washing and cooling with cold tapwater. The water turns off automatically when the required temperature is reached. Should the cooling unit fail, **DO NOT IMMERSE THE BOMB IN WATER.** The bomb may be held inverted over the sink with a piece of tubing connected to the tap flows water into the bomb from below.

The water emerging from the Cooler will be contaminated with acids and ash produced by the combustion of the analysed samples **and must be discarded**.

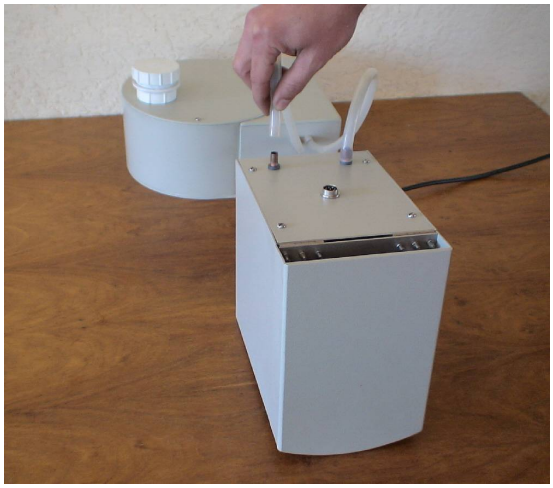
INSTALLATION

Hold the connecting pipes upwards while filling the Constant Temperature Tank with good quality tap water to the BOTTOM OF THE FILLER SPOUT. (Use de-ionised water if your water supply contains high lime or sediment content.)

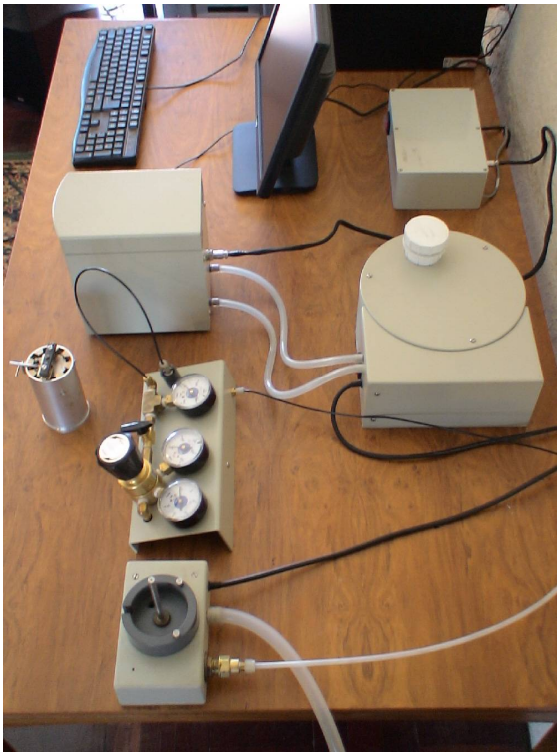


Place the Tank at the rear of the workbench, and the Calorimeter in front of it, **with it's back panel upwards**. Keeping the pipes upwards, attatch the connecting pipes onto the Calorimeter as shown, without crossing the pipes over.

If you are connecting up a double Drycal system, please refer to the notes on the next page

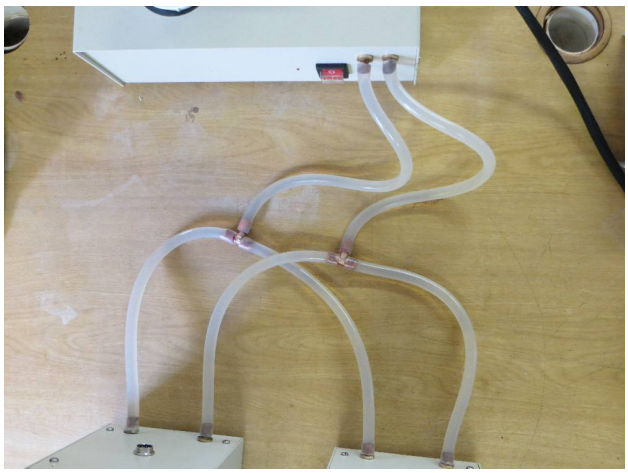


Place the other items as shown in the picture, with the Bomb Cooler/Washer on the right.
The large diameter drain pipe must go to waste, such as into a sink or bucket. NB. This pipe must slope downwards, otherwise the Cooler will overflow. Connect the 1/4" pipe to the cold water supply.
Connect the Oxygen Charging Station to the oxygen supply using the supplied 1/8" tubing. It may be connected directly to the bottle without regulator, provided that copper or stainless piping is used.
Connect the calorimeter to the Controller box using the 8 pin connector cable, and to the PC using the attatched Network Cable.
Connect screen, mouse and keyboard.
Connect all mains cables via the supplied multiway plug extender.
The above layout can be reversed if required.



CONNECTION OF DOUBLE CALORIMETER SYSTEM.

When two Drycal Calorimeter Modules are to be used in a system, the water piping from the Constant Temperature Tank and the calorimeters must be connected as shown below. **Cal 1 must be on the LEFT HAND SIDE.**



The system can optimally be run using three bombs, each bomb can go into **either** calorimeter. Normally the operator should charge a bomb and load it into Cal 1. He should then charge the next bomb and load it into Cal 2, following which he should charge the next bomb to have it ready to load it into Cal 1 as soon as the first analysis is completed.

PREPARING FOR USE.

1. Turn on water and oxygen supplies to the calorimeter. Make sure the pressure at the Oxygen Charging Station regulator is set to 2500 kPa. Do not use faulty regulators or gauges, since this can result in over pressurising the bomb.

2. Switch on the PC controller, Calorimeter Interface and Constant Temperature Tank. The latter provides thermostatically controlled water (at about 36 degrees Centigrade) that flows through the Outer Jacket of the Calorimeter(s), providing a stable temperature environment around the Bomb whilst the determination is in progress.

3. Start the Drycal program by double clicking on the "Shortcut to DRYCAL 2010 icon

4. If the daily cleaning procedure was not done on the bomb(s) at the close of the last day or shift it must be done now. See Daily Maintenance on page 11.

5. Twenty minutes after switch on, the calorimeter is ready for use.

ANALYSING A SAMPLE

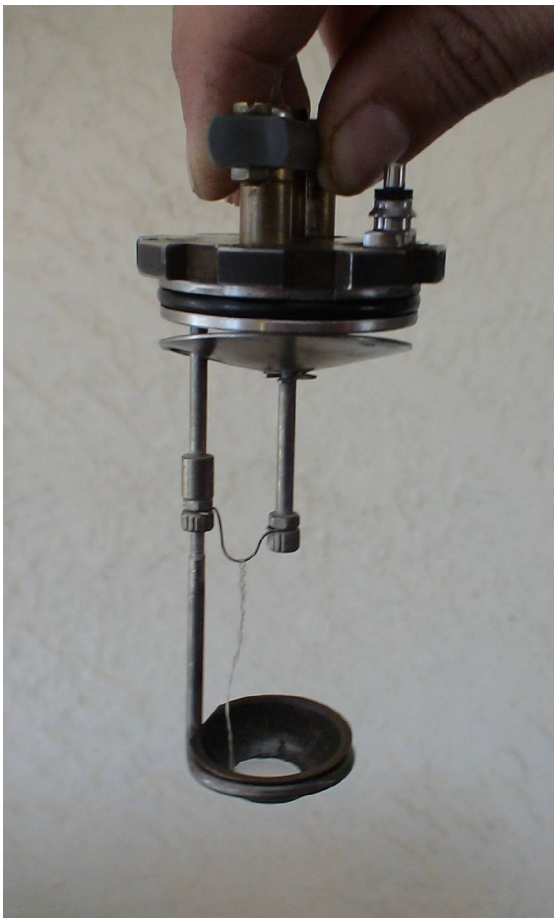


Make a small "ball" at the end of a 150 mm length of firing cotton, and put this ball into a clean crucible (supplied). Place the crucible onto your balance pan, with the protruding length of cotton also resting on the pan, and TARE the balance to zero.

Weigh in the required quantity of sample (for coal samples and Benzoic acid use 0.5 g, for oil samples use 0.2 g)

Press the F2 key, and type the sample ID into the window that appears on the screen. **Type "CAL" as the ID if you are weighing up Benzoic Acid or other calibration material.** Press the Tab key, and type in the sample mass. (Only weights from 0.2 to 0.6 gram may be used)

If the sample is of low combustibility, such as a discard, sink or ash, it may be "**spiked**" with a material of high combustibility, such as powdered Benzoic Acid, to assist complete ignition. Press the Tab key to bring the cursor into the Spike Weight box and enter the mass of spiking material (such as 0.2 gram). Check that the CV of the spiking material is correctly displayed, if not the correct CV must be entered after pressing the F9 key. Use a glass rod to mix the sample and spiking agent.



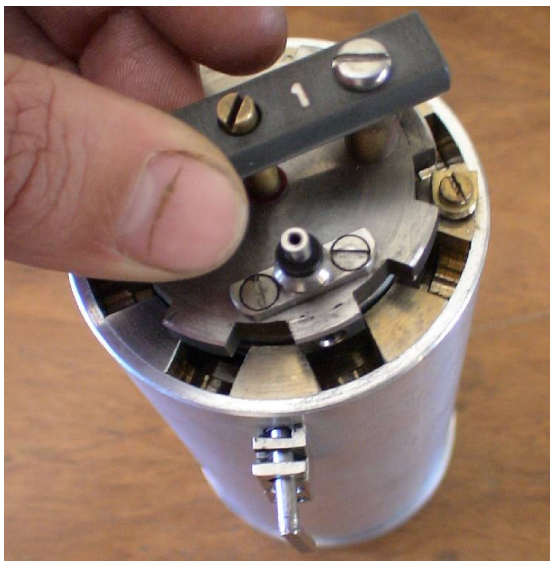
Remove the crucible from the balance, and gently tap it down a few times on the bench to compact the loose sample slightly. This captures the cotton, and promotes better burning. **Some low CV materials are best determined by using a reduced sample mass and not spiking.**

Hold the Bombcap with the electrodes downwards and place the crucible onto the support ring in the longer electrode with the flat on the edge of the crucible adjacent to the electrode.

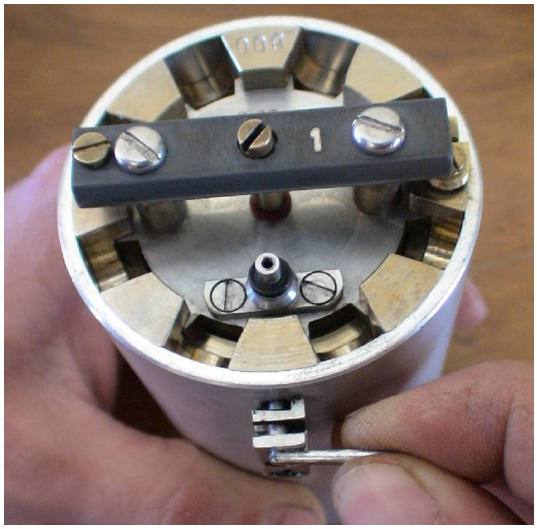
Lift the end of the cotton over the firing wire, and fold the last +/- 15 mm of the cotton back against itself.

Roll the threads together so as to attatch them to the firing wire. It helps to pull slightly downwards as the fingers are taken off the cotton.

With the Oxygen Valvepin pulled fully outwards, lower the cap carefully into the Bomb with the Oxygen Filling Post in the 5 o'clock position. Ease the Bombcap gently downwards with a slight rocking motion, press it down as far as it will go. Lock the cap with a clockwise twist, as far as it will go.



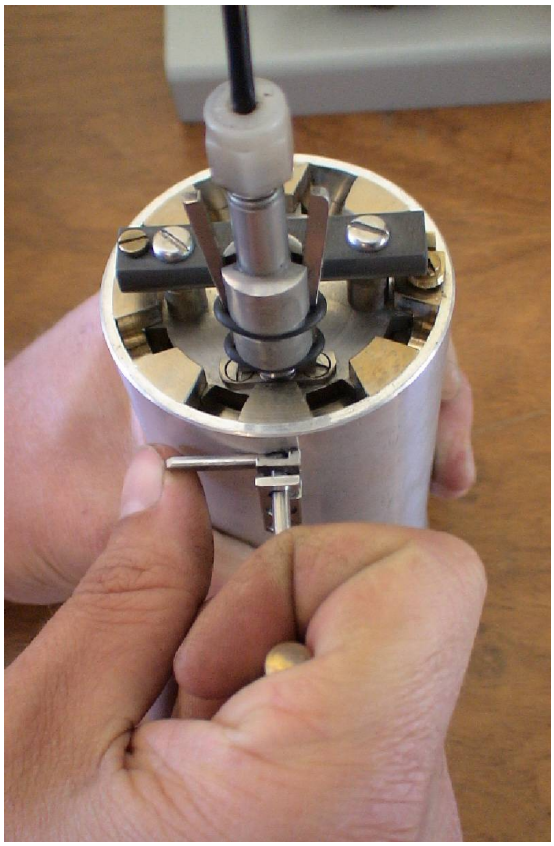
Hold the Valvepin lever in the 3 o'clock position, and move it inwards so that it enters into the Locking Block.



Rotate the lever counter clockwise to 3 o'clock. Attach the Oxygen Filler by pressing it down onto the Filling Post, and open the tap on the Oxygen Filling Station.



When the pressure indicated on the left gauge has risen to 25 bar, press the Valve Pin inwards using the supplied T shaped tool.



Lift the valvepin to the 12 o'clock position, which closes off the bomb. Oxygen will be heard escaping, this is quite normal. Turn off the tap on the Filling Station, then remove the Oxygen Filler by pressing inwards on the two levers.



Holding the Bomb by the hanle on the cap, lower the Bomb carefully into the calorimeter with Locking Block facing forward, and close the lid firmly. The analysis will start, and continue automatically through to completion, with messages displayed on the monitor, giving current status, jacket and bomb temperatures.



WHILE THE ANALYSIS IS IN PROGRESS, AS A STANDARD SAFETY PRECAUTION, AVOID HAVING ANY PART OF YOUR BODY ABOVE THE CALORIMETER.

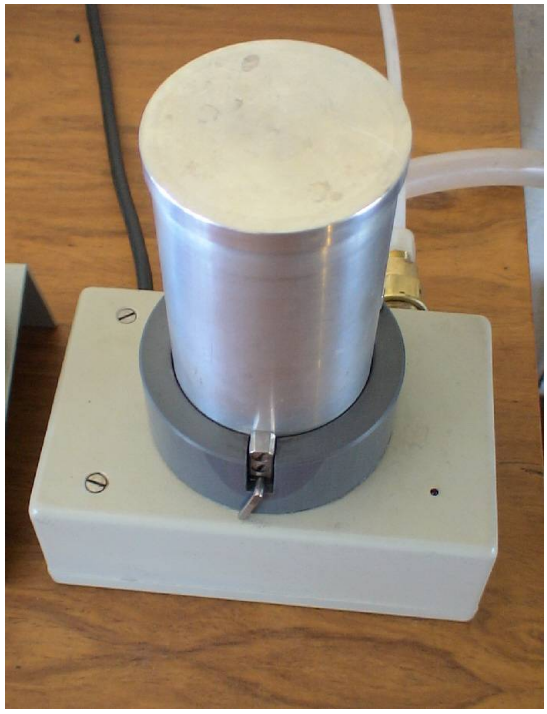
When the analysis cycle terminates, the result is displayed. Press the release button on the front of the lid, and open the lid fully. Remove the Bomb from the calorimeter.

If your system includes a second bomb, it can now be loaded into the Calorimeter and the next analysis started.

Move the Valvepin from 12 o'clock to 9 o'clock, and allow the pressure to escape from the Bomb. When the outrush of gas ceases, rotate the Valvepin through the 12 o'clock position to 3 o'clock, and pull the pin outwards as far as it will go. Rotate the Bombcap counter-clockwise until the lugs on the cap align with the gaps of the bomb, then ease the cap out of the bomb.

Look down into the Bomb to check that no coal was spilt when the sample just analysed was loaded. Discard the result if ANY coal can be seen.

Place the Bomb upside down onto the Bomb Cooling Station. A red light will come on to indicate that cooling and washing of the Bomb is in progress.



While the Bomb cools, place the Bombcap into the supplied vessel to remove contamination and any residual heat from the assembly.

The water level must be maintained near the top of the vessel, so as to submerge the large O-Ring on the Bombcap, and should be changed whenever the water becomes cloudy.

When the red light on the Bomb Cooler goes off, lift the bomb off the Cooler and shake it to remove excess water.

Lift the Bombcap out of the vessel and shake off excess water. Dry off the electrodes and firing wire, before loading the next sample.



Periodic check determinations should be run on Benzoic Acid **and** a good Control Standard. If the result obtained deviates from the correct value, refer to the CALIBRATION section.

NB. The CV. of coal standards <u>reduces with time</u>.

OVERNIGHT, following a thorough cleaning, switch off ALL itens at the mains switch. This will prevent damage from lightning strikes and other electrical surges while the instrument is not in use. Turn off water and oxygen supplies.

CALIBRATION.

NB : Re-calibration should only be required infrequently, such as following the replacement of some major component of the calorimeter. If the result obtained on Benzoic Acid or your control standard deviates from the correct value, first check that this is not being caused by leaking seals.

1. Perform determinations on Benzoic Acid (or other calibration standard) until you have obtained sufficient results that are in good agreement with each other to satisfy calibration requirements, at least two results must be included. NB These deperminations must have been entered with "CAL" as the sample ID.

2. Press F8 on the keyboard. You will be asked which bomb you wish to calibrate. Type in the Bomb ID number and press Enter. A table of the last five determinations having "CAL" as ID that were done in this Bomb will appear on the screen. Press the letter of the alphabet that appears alongside each of the calibration determinations that you wish to include. DO NOT INCLUDE ANY RESULT THAT DOES NOT CHECK WITH THE OTHER SELECTED RESULTS. Press ENTER when complete. A red mark will appear to the right of each determination that has been used in a calibration sequence.

3. A window will appear advising on any deletions that were necessary for reasons of poor agreement. Clear this window with the ESC key, as well as the next window, that advises on the success of the calibration.

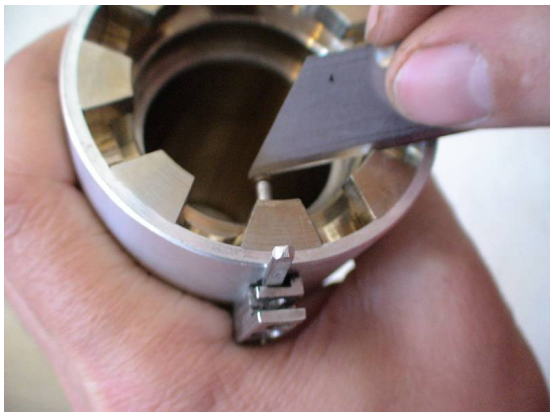
If a Coal Standard is used, remember that it's Calorific Value always DECREASES with time, and it's current value must be established on an on-going basis, using Benzoic Acid to calibrate the instrument first. Even if an instrument is always calibrated on Benzoic Acid, it is essential to run regular coal standards to fully monitor correct calorimeter operation. Some fault conditions, such as low oxygen pressure at firing, can produce a normal result on Benzoic Acid, but a LOW value on a coal standard, where the layer of ash may prevent total combustion.

ROUTINE MAINTENANCE

DAILY MAINTENANCE (or AFTER EVERY 50 FIRINGS)

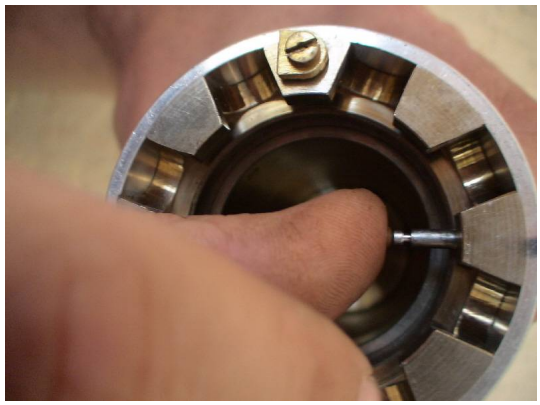
Use detergent and a toothbrush to scrub all residues off the Bomb, Bombcap and electrodes

Inspect the large 2-221 O-Ring on the Bombcap, and replace as shown below if damaged or burnt. Grease lightly if OK.



Remove the 2-002 O-Ring from the Valvepin by cutting it off as shown

Lightly grease a new 2-002 Viton O-Ring and fit by pressing it onto the end of the Valvepin Shaft. (Note that the Valvepin can only be removed from the Bomb once the O-Ring has been removed)



Fit a new firing wire. Use fine water paper to remove any corrosion on the electrodes.

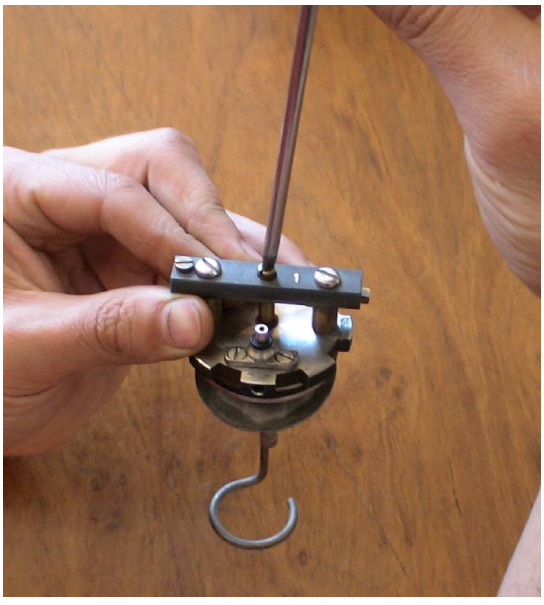
CAUTION

EXCESS GREASE IS LIABLE TO IGNITE AND MAY SET FIRE TO THE O-RING !

AFTER EVERY FIVE HUNDRED FIRINGS

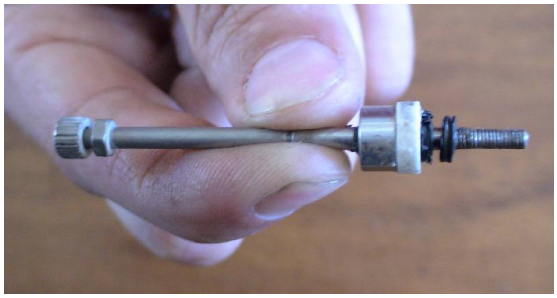
Remove the Firing Wire.

Unscrew the Electrode Contactor as shown, and remove the Live Electrode from the Bombcap. Carefully clean the bore through the centre of the Bombcap.



NB Don't loose the small red insulating washer that sits on the top of the Bombcap, under the Contactor !

Thoroughly clean the Electrode, and refit with TWO new Q-006 Quad Seals as shown.



Refit into the Bombcap, replace the red insulating washer, and screw down the Contactor. (DO NOT OVERTIGHTEN)

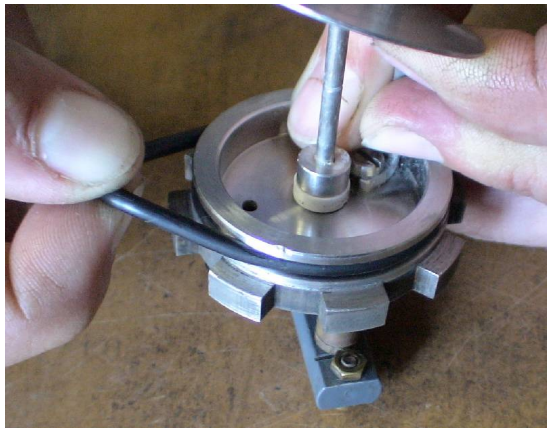
Remove the 2-221 O-Ring from the Bombcap as shown, and discard.



Lightly grease a new 2-221 O-Ring and fit as shown.

NB. The O-Ring must NOT be rolled into the groove, since this will cause it to be fitted with a twist.

Fit the ring in on one side, then pull it outwards and let it fall back into the groove



FAULT CONDITION MESSAGES

The following messages will appear when the listed fault conditions arise. They are given, as far as is possible, in the same order that they could occur.

"MISFIRE IN CAL X"

If the cotton has burned, but the sample is completely unburnt:

1. The cotton may have pulled out from under the sample, or
2. A drop of water fell onto the cotton.

If the sample has partially burnt:

1. The 2-002 O-Ring on the Fill Pin may be damaged.

"ANALOGUE OVERFLOW IN CAL X"

This fault condition occurs if, after combusting the sample, the Bomb Temperature rises to the limit of the A/D convertor. Possibly the Bomb was not sufficiently cooled at the end of the preceeding analysis, otherwise the mass of sample was too high.

SERVICE ADVICE BY PHONE

If you encounter a problem that you are unable to rectify after referring to the information in this OPERATORS MANUAL, seek further advice from your supervisor.

DO NOT ATTEMPT TO DISMANTLE, REPAIR OR ADJUST ANY PART OF THE INSTRUMENT UNLESS SPECIFICALLY INSTRUCTED TO IN THE MANUALS. YOU WILL PROBABLY BE CREATING GREATER PROBLEMS BY DOING SO.

If the problem persists after a careful study of the manual, seek advice from your Local Agent, or from :

ENERGY INSTRUMENTATION (PTY) LTD

Telephone	076 315 9031
	079 893 5103
e-mail	mail@energyinst.co.za

A SHORT EXPLANATION OF CALORIMETRY.

Bomb Calorimetry determines the amount of heat that is produced when a specific weight of a material, such as a coal, is burnt.

This quantity is known as the ***Calorific Value***, and is the **most important property** of a fuel material. It determines, for example, how much electricity a ton of this coal can produce, or how many bricks it can bake. For this reason, contracts between buyers and sellers of coal usually centre around the Calorific Value, which is why these determinations must be done with great care, so as to provide results that are accurate and dependable.

The method of determining this heat release is very simple. We weigh an amount of the material into a special heat resisting crucible, which we load into a strong container, known as the "bomb". We then pressurise this bomb with oxygen, and burn the material, measuring the rise in temperature that this produces in the bomb assembly.

This temperature rise is used by the calorimeter controller to calculate the C.V., taking the weight of sample into account, as well as the "calibration factor" and a few other things.

As you can imagine, when the coal burns and the bomb gets hot, some of this heat is lost to the surroundings. Obviously more heat will be lost on a cold day than on a hot day, and this would cause variable results.

In order to prevent this, so that the result produced is the same in a variety of room temperatures, we surround the bomb with an outer jacket that has water at a constant temperature circulated through it. This reduces the effects of the room temperature on the determination.

Good cleanliness is essential in calorimetry, to ensure a high level of safety. The pressure of the oxygen in the bomb is very high, and a leak of burning gas after the sample ignites can cause a very dramatic fire to emerge that can burn the bomb, rather like an oxy-acetylene cutting torch. Take care of your instrument as described in the maintenance schedules so that this won't happen to you!